

Vulnerability Validation Report

CVE-2026-46331 / pedit COW

Linux kernel act_pedit - local privilege escalation and auditd/AppArmor detection

Field	Value
Test system	Ubuntu 24.04.4 LTS, kernel 6.17.0-35-generic
Host	jce, VirtualBox virtual machine
Test date	June 30, 2026
Scope	Vulnerability validation, artifact collection, mitigation and detection
Test author	Matek Statek
Classification	Laboratory / defensive material

Note: this report describes validation performed in a controlled laboratory environment. It is not an instruction for exploiting the vulnerability in production environments or against third-party systems.

1. Executive Summary

The practical impact of CVE-2026-46331, known as pedit COW, was confirmed in a laboratory environment. The tested host met the conditions required to exploit local privilege escalation: the `act_pedit` module was available, unprivileged user namespaces were enabled, and `kernel.apparmor_restrict_unprivileged_unconfined` had a value of 0 before mitigation.

Before mitigation, `auditd` registered a sequence of events leading from the `packet_edit_meme` process, executed by an unprivileged user, to the execution of `/usr/bin/su`, `/usr/bin/dash`, `/usr/bin/whoami` and `/usr/bin/id` with root privileges. In parallel, the kernel journal registered `tc action pedit offset out of bounds` messages, time-correlated with the exploitation attempt.

After enabling `kernel.apparmor_restrict_unprivileged_unconfined=1`, the repeated attempt did not result in analogous execution of processes with `uid=0` in the context of the tested variant. The logs instead showed AppArmor denials for the `sys_admin` and `net_admin` capabilities in the `unprivileged_usersns` profile. Conclusion: the applied mitigation limited the tested escalation path, but it does not replace a kernel update.

Area	Finding
Pre-mitigation result	Confirmed escalation to root in the lab environment
Post-mitigation result	No analogous <code>root_exec</code> sequence for <code>su/dash/whoami/id</code> ; AppArmor denials visible
Key detections	<code>ns_create</code> , <code>netlink_socket</code> , <code>apparmor_bypass_path</code> , <code>root_exec</code> , <code>tc action pedit offset out of bounds</code> , AppArmor DENIED <code>sys_admin/net_admin</code>
Audit reliability	<code>auditd: lost=0, backlog_limit=32768, rate_limit=0</code>

2. Scope and Test Objective

The objective of the test was to prepare evidentiary material for a validation report: to check whether the host met the vulnerability conditions, confirm the effect before mitigation, implement the AppArmor mitigation and compare system behavior after mitigation. Particular emphasis was placed on detection and artifacts that can be used in a SOC, SIEM or vulnerability management process.

The scope covered one Ubuntu 24.04.4 LTS laboratory host running as a virtual machine. The test concerned local privilege escalation and required the ability to execute code on the host. No remote attack from the Internet was tested.

Scope element	Description
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Validation of vulnerability conditions	Checking the kernel, act_pedit, user namespaces, AppArmor and unconfined profiles
Behavior audit	auditd, aureport, ausearch, kernel journal, dmesg, AppArmor
Before/after comparison	Pre-mitigation state compared with the state after kernel.apparmor_restrict_unprivileged_unconfined=1
Detection	auditd rules, kernel journal indicators, event-sequence correlation

3. Vulnerability Overview and Exploitation Conditions

CVE-2026-46331 affects the net/sched/act_pedit mechanism in the Linux kernel. The NVD and Ubuntu descriptions indicate a partial COW bug leading to page cache corruption. The issue results from calculating the copy-on-write range before the key loop in tcf_pedit_act(), without considering the runtime header offset added by typed keys. As a result, part of the write region may remain without proper COW handling, which can consequently corrupt shared page cache memory.

A practical exploitation scenario requires local code execution. In the tested variant, the important factors were: the ability to create user namespaces, the availability of act_pedit, and an AppArmor/aa-exec path to a profile that allowed the required capability inside the namespace. The attack is not a classic Internet-facing RCE, but it is significant for systems where a user, application, webshell, CI/CD runner or low-privileged process can execute code locally.

Parameter	Value
Vulnerability type	Local Privilege Escalation
Component	Linux kernel net/sched/act_pedit
Mechanism	Partial COW / page cache corruption
Required access	Local code execution as an unprivileged user
Required conditions in the tested variant	unprivileged user namespaces, act_pedit, netlink/tc, aa-exec/AppArmor path
Possible impact	Execution of processes as root

4. Test Environment

The following data comes from auditd evidence bundles collected after the test. The environment was an Ubuntu 24.04.4 LTS virtual machine with kernel 6.17.0-35-generic.

Extract 1. Host and operating system data

```
Static hostname: jce
                Chassis: vm
  Virtualization: oracle
Operating System: Ubuntu 24.04.4 LTS
                Kernel: Linux 6.17.0-35-generic
                Architecture: x86-64
  Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
PRETTY_NAME="Ubuntu 24.04.4 LTS"
VERSION_ID="24.04"
Linux jce 6.17.0-35-generic #35~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue May 26
19:30:42 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
```

Extract 2. Kernel packages

```
ii  linux-headers-6.17.0-35-generic      6.17.0-35.35~24.04.1
amd64 Linux kernel headers for version 6.17.0
ii  linux-image-6.17.0-35-generic        6.17.0-35.35~24.04.1
amd64 Signed kernel image generic
ii  linux-modules-6.17.0-35-generic      6.17.0-35.35~24.04.1
amd64 Linux kernel extra modules for version 6.17.0
ii  linux-modules-extra-6.17.0-35-generic 6.17.0-35.35~24.04.1
amd64 Linux kernel extra modules for version 6.17.0
linux-image-6.17.0-35-generic:
  Installed: 6.17.0-35.35~24.04.1
  Candidate: 6.17.0-35.35~24.04.1
```

Extract 3. act_pedit availability and related modules

```
filename:      /lib/modules/6.17.0-35-generic/kernel/net/sched/act_pedit.ko.zst
description:   Generic Packet Editor actions
alias:         net-act-pedit
depends:
intree:        Y
name:          act_pedit
vermagic:      6.17.0-35-generic SMP preempt mod_unload modversions
act_pedit      20480  0
cls_basic      12288  0
sch_ingress    20480  0
```

Extract 4. tc / iproute2

```
/usr/sbin/tc
tc utility, iproute2-6.1.0, libbpf 1.3.0
ii  iproute2                                6.1.0-1ubuntu6.2
amd64 networking and traffic control tools
```

The state of AppArmor mechanisms and user namespaces was critical for risk assessment. Before mitigation, kernel.apparmor_restrict_unprivileged_unconfined had a value of 0. After mitigation, the setting was changed to 1.

Parameter	Before mitigation	After mitigation
kernel.unprivileged_userns_clone	1	1

user.max_user_namespaces	30124	30124
kernel.apparmor_restrict_unprivileged_userns	1	1
kernel.apparmor_restrict_unprivileged_unconfined	0	1
act_pedit	loaded, Used by=1	loaded, Used by=0
auditd lost	0	0

Extract 5. AppArmor - profiles and relevant unconfined profiles

```

apparmor module is loaded.
153 profiles are loaded.
56 profiles are in enforce mode.
5 profiles are in complain mode.
92 profiles are in unconfined mode.
32 processes have profiles defined.
2 processes are unconfined but have a profile defined.
  busybox
  chrome
  crun
  firefox
  flatpak
  lxc-unshare
  lxc-userns
  podman
  rootlesskit
  runc
  slirp4netns
  trinity

```

5. Audit Methodology and Evidence Collection

Auditd was configured to record events characteristic of local privilege escalation: namespace creation and switching, netlink socket usage, kernel module loading, and executions of processes with `euid=0` by an unprivileged user. Additional watches were enabled on tools relevant to this vulnerability class: `aa-exec`, `unshare`, `nsenter`, `tc`, `modprobe`, `kmod`, `su`, `sudo` and shells.

Auditd does not directly record page cache corruption or kernel memory modification. It does, however, register actions that prepare the exploitation path and the final effect in the form of processes executed with `euid=0`. For this reason, auditd logs were supplemented with the kernel journal, where `tc` action `pedit` offset out of bounds messages were registered.

Extract 6. Loaded auditd rules

```

-a always,exit -F arch=b64 -S clone,unshare,setns,clone3 -F auid>=1000 -F auid!=-1 -F key=ns_create
-a always,exit -F arch=b64 -S init_module,delete_module,finit_module -F auid>=1000 -F auid!=-1 -F key=kernel_module
-a always,exit -F arch=b64 -S socket -F a0=0x10 -F auid>=1000 -F auid!=-1 -F key=netlink_socket
-a always,exit -F arch=b64 -S execve -F euid=0 -F auid>=1000 -F auid!=-1 -F key=root_exec

```

```
-w /usr/bin/aa-exec -p x -k apparmor_bypass_path
-w /usr/bin/unshare -p x -k ns_tools
-w /usr/bin/nsenter -p x -k ns_tools
-w /usr/sbin/tc -p x -k tc_exec
-w /sbin/tc -p x -k tc_exec
-w /usr/sbin/modprobe -p x -k modprobe_exec
-w /sbin/modprobe -p x -k modprobe_exec
-w /usr/bin/kmod -p x -k modprobe_exec
-w /usr/bin/su -p x -k suid_exec
-w /usr/bin/sudo -p x -k suid_exec
-w /bin/bash -p x -k shell_exec
-w /bin/sh -p x -k shell_exec
```

Extract 7. auditd status before mitigation

```
enabled 1
failure 1
pid 65375
rate_limit 0
backlog_limit 32768
lost 0
backlog 2
backlog_wait_time 60000
backlog_wait_time_actual 0
loginuid_immutable 0 unlocked
```

Extract 8. auditd status after mitigation

```
enabled 1
failure 1
pid 65375
rate_limit 0
backlog_limit 32768
lost 0
backlog 0
backlog_wait_time 60000
backlog_wait_time_actual 0
loginuid_immutable 0 unlocked
```

In both phases, the lost counter had a value of 0, which means that kernel audit did not report any lost events. The backlog value was 2 before mitigation and 0 after mitigation. The backlog limit was set to 32768 and rate_limit to 0 to reduce the risk of cutting events during the test.

6. Pre-Mitigation Test Results

Before mitigation, the host met the conditions that favored exploitation. The kernel.apparmor_restrict_unprivileged_unconfined value was 0, act_pedit was available and, after the test was started, it was used by the traffic control mechanism. Auditd registered a sequence typical of escalation: ns_create, netlink_socket, aa-exec to the trinity profile, and root_exec for su, dash, whoami and id.

Extract 9. Sysctl before mitigation

```
kernel.unprivileged_usersns_clone = 1
user.max_user_namespaces = 30124
```

```
kernel.apparmor_restrict_unprivileged_userns = 1
kernel.apparmor_restrict_unprivileged_unconfined = 0
```

Extract 10. Aureport - pre-mitigation event sequence

```
605. 06/30/2026 17:34:44 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2672
606. 06/30/2026 17:34:44 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2675
607. 06/30/2026 17:34:44 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2677
608. 06/30/2026 17:34:44 apparmor_bypass_path yes /usr/bin/aa-exec user 2678
609. 06/30/2026 17:34:44 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2679
610. 06/30/2026 17:34:44 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2680
611. 06/30/2026 17:34:44 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2681
612. 06/30/2026 17:34:45 root_exec yes /usr/bin/su user 2682
613. 06/30/2026 17:34:45 root_exec yes /usr/bin/dash user 2683
614. 06/30/2026 17:34:50 root_exec yes /usr/bin/whoami user 2684
615. 06/30/2026 17:34:53 root_exec yes /usr/bin/id user 2685
616. 06/30/2026 17:34:56 root_exec yes /usr/bin/ls user 2686
```

Extract 11. Kernel journal - pre-mitigation pedit messages

```
Jun 30 17:34:45 jce kernel: tc action pedit offset 17682 out of bounds
Jun 30 17:34:45 jce kernel: tc action pedit offset 17718 out of bounds
Jun 30 17:34:46 jce kernel: tc action pedit offset 17718 out of bounds
Jun 30 17:34:47 jce kernel: tc action pedit offset 17718 out of bounds
Jun 30 17:34:51 jce kernel: tc action pedit offset 17718 out of bounds
Jun 30 17:34:58 jce kernel: tc action pedit offset 17718 out of bounds
Jun 30 17:35:11 jce kernel: tc action pedit offset 17718 out of bounds
Jun 30 17:35:37 jce kernel: tc action pedit offset 17718 out of bounds
```

The full audit.log lines for the most important events are included below. The lines were not shortened or replaced with ellipses. Event 2678 shows the use of aa-exec with the trinity profile, while events 2682-2685 show su, dash, whoami and id executed with root privileges with auid=1000.

Extract 12. audit.log - full blocks of key pre-mitigation events

```
===== audit event 2678 =====
type=SYSCALL msg=audit(1782840884.588:2678): arch=c000003e syscall=59 success=yes
exit=0 a0=7ffe299084c0 a1=7ffe299085e0 a2=7ffe29908920 a3=48d02b items=2 ppid=66971
pid=66972 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000
sgid=1000 fsgid=1000 tty=pts0 ses=3 comm="aa-exec" exe="/usr/bin/aa-exec"
subj=unconfined key="apparmor_bypass_path"
type=EXECVE msg=audit(1782840884.588:2678): argc=6 a0="aa-exec" a1="-p"
a2="trinity" a3="--" a4="./packet_edit_meme" a5="--in-profile"
type=CWD msg=audit(1782840884.588:2678):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840884.588:2678): item=0 name="/usr/bin/aa-exec"
inode=524801 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1782840884.588:2678): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
```

```

cap_fp=0 cap-fi=0 cap-fe=0 cap-fver=0 cap-frootid=0
type=PROCTITLE msg=audit(1782840884.588:2678):
proctitle=61612D65786563002D70007472696E697479002D2D002E2F7061636B65745F656469745F6
D656D65002D2D696E2D70726F66696C65
===== audit event 2682 =====
type=SYSCALL msg=audit(1782840885.380:2682): arch=c000003e syscall=59 success=yes
exit=0 a0=48d0d6 a1=7fffd1970310 a2=0 a3=0 items=2 ppid=66971 pid=66972 auid=1000
uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=1000 sgid=1000 fsgid=1000 tty=pts0
ses=3 comm="su" exe="/usr/bin/su" subj=trinity key="root_exec"
type=EXECVE msg=audit(1782840885.380:2682): argc=1 a0="/bin/su"
type=CWD msg=audit(1782840885.380:2682):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840885.380:2682): item=0 name="/bin/su" inode=582897
dev=08:02 mode=0104755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap-fi=0
cap-fe=0 cap-fver=0 cap-frootid=0
type=PATH msg=audit(1782840885.380:2682): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap-fi=0 cap-fe=0 cap-fver=0 cap-frootid=0
type=PROCTITLE msg=audit(1782840885.380:2682):
proctitle=61612D65786563002D70007472696E697479002D2D002E2F7061636B65745F656469745F6
D656D65002D2D696E2D70726F66696C65
===== audit event 2683 =====
type=SYSCALL msg=audit(1782840885.392:2683): arch=c000003e syscall=59 success=yes
exit=0 a0=7fff3998e428 a1=7fff3998e418 a2=0 a3=0 items=2 ppid=66971 pid=66972
auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts0 ses=3
comm="sh" exe="/usr/bin/dash" subj=trinity key="root_exec"
type=EXECVE msg=audit(1782840885.392:2683): argc=1 a0="/bin/sh"
type=CWD msg=audit(1782840885.392:2683):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840885.392:2683): item=0 name="/bin/sh" inode=524930
dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap-fi=0
cap-fe=0 cap-fver=0 cap-frootid=0
type=PATH msg=audit(1782840885.392:2683): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap-fi=0 cap-fe=0 cap-fver=0 cap-frootid=0
type=PROCTITLE msg=audit(1782840885.392:2683):
proctitle=61612D65786563002D70007472696E697479002D2D002E2F7061636B65745F656469745F6
D656D65002D2D696E2D70726F66696C65
===== audit event 2684 =====
type=SYSCALL msg=audit(1782840890.852:2684): arch=c000003e syscall=59 success=yes
exit=0 a0=56e004011530 a1=56e004011510 a2=56e004011520 a3=8 items=2 ppid=66972
pid=66993 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0
tty=pts0 ses=3 comm="whoami" exe="/usr/bin/whoami" subj=trinity key="root_exec"
type=EXECVE msg=audit(1782840890.852:2684): argc=1 a0="whoami"
type=CWD msg=audit(1782840890.852:2684):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840890.852:2684): item=0 name="/usr/bin/whoami"
inode=525835 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap-fi=0 cap-fe=0 cap-fver=0 cap-frootid=0
type=PATH msg=audit(1782840890.852:2684): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap-fi=0 cap-fe=0 cap-fver=0 cap-frootid=0
type=PROCTITLE msg=audit(1782840890.852:2684): proctitle="whoami"
===== audit event 2685 =====
type=SYSCALL msg=audit(1782840893.402:2685): arch=c000003e syscall=59 success=yes
exit=0 a0=56e004011530 a1=56e004011510 a2=56e004011520 a3=8 items=2 ppid=66972
pid=67011 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0
tty=pts0 ses=3 comm="id" exe="/usr/bin/id" subj=trinity key="root_exec"
type=EXECVE msg=audit(1782840893.402:2685): argc=1 a0="id"

```

```
type=CWD msg=audit(1782840893.402:2685):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840893.402:2685): item=0 name="/usr/bin/id" inode=525215
dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0
cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1782840893.402:2685): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840893.402:2685): proctitle="id"
```

Conclusion from the pre-mitigation phase: the vulnerability was practically confirmed in the laboratory environment. An unprivileged user caused processes to execute with uid/euid=0. The correlation of auditd and kernel journal indicates a path involving user namespaces, aa-exec, netlink and act_pedit.

7. Mitigation and Post-Mitigation Results

After the phase confirming the vulnerability, AppArmor mitigation was enabled by setting `kernel.apparmor_restrict_unprivileged_unconfined=1`. This setting limits the ability of an unprivileged unconfined process to use aa-exec to switch to a more favorable AppArmor profile. The mitigation was applied without disabling `kernel.unprivileged_usersns_clone`, so the test specifically checked the impact of additional AppArmor hardening.

Extract 13. Sysctl after mitigation

```
kernel.unprivileged_usersns_clone = 1
user.max_user_namespaces = 30124
kernel.apparmor_restrict_unprivileged_usersns = 1
kernel.apparmor_restrict_unprivileged_unconfined = 1
```

After mitigation, attempts were still visible in auditd. `ns_create`, `netlink_socket` and subsequent uses of aa-exec with the trinity, chrome and flatpak profiles were registered. However, no analogous `root_exec` sequence for `/usr/bin/su`, `/usr/bin/dash`, `/usr/bin/whoami` or `/usr/bin/id` was observed in the context of the post-mitigation attempt.

Extract 14. Aureport - post-mitigation events related to the attempt

```
813. 06/30/2026 17:36:11 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2984
814. 06/30/2026 17:36:11 apparmor_bypass_path yes /usr/bin/aa-exec user 2985
815. 06/30/2026 17:36:11 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2987
816. 06/30/2026 17:36:11 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2990
817. 06/30/2026 17:36:11 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2991
818. 06/30/2026 17:36:11 apparmor_bypass_path yes /usr/bin/aa-exec user 2992
819. 06/30/2026 17:36:11 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2994
820. 06/30/2026 17:36:11 netlink_socket yes /home/user/Downloads/packet_edit_meme-
```

```

main/packet_edit_meme user 2997
821. 06/30/2026 17:36:11 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 2998
822. 06/30/2026 17:36:11 apparmor_bypass_path yes /usr/bin/aa-exec user 2999
823. 06/30/2026 17:36:11 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3001
824. 06/30/2026 17:36:11 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3004
825. 06/30/2026 17:36:31 ns_create yes /snap/firefox/7766/usr/lib/firefox/firefox
user 3012
826. 06/30/2026 17:36:37 netlink_socket yes
/snap/firefox/7766/usr/lib/firefox/firefox user 3015
827. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3020
828. 06/30/2026 17:36:54 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3023
829. 06/30/2026 17:36:54 apparmor_bypass_path yes /usr/bin/aa-exec user 3025
830. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3026
831. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3029
832. 06/30/2026 17:36:54 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3031
833. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3032
834. 06/30/2026 17:36:54 apparmor_bypass_path yes /usr/bin/aa-exec user 3033
835. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3035
836. 06/30/2026 17:36:54 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3038
837. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3039
838. 06/30/2026 17:36:54 apparmor_bypass_path yes /usr/bin/aa-exec user 3040
839. 06/30/2026 17:36:54 ns_create yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3042
840. 06/30/2026 17:36:54 netlink_socket yes /home/user/Downloads/packet_edit_meme-
main/packet_edit_meme user 3045

```

The key difference after mitigation was the presence of AppArmor DENIED entries for the `sys_admin` and `net_admin` capabilities. They indicate that the process attempted to perform operations required to prepare the path, but was restricted by the `unprivileged_usersns` profile and the profile stacking mechanism.

Extract 15. audit.log - full blocks of key post-mitigation events

```

===== audit event 2981 =====
type=AVC msg=audit(1782840971.321:2981): apparmor="DENIED" operation="capable"
class="cap" profile="unprivileged_usersns" pid=67391 comm="packet_edit_mem"
capability=21 capname="sys_admin"
type=SYSCALL msg=audit(1782840971.321:2981): arch=c000003e syscall=257 success=no
exit=-13 a0=ffffff9c a1=48d05a a2=1 a3=0 items=1 ppid=67390 pid=67391 auid=1000
uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000
tty=pts0 ses=3 comm="packet_edit_mem" exe="/home/user/Downloads/packet_edit_meme-
main/packet_edit_meme" subj=unprivileged_usersns key=(null)
type=CWD msg=audit(1782840971.321:2981):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840971.321:2981): item=0 name="/proc/self/setgroups"

```

```

inode=287679 dev=00:17 mode=0100644 ouid=1000 ogid=1000 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840971.321:2981): proctitle="./packet_edit_meme"
===== audit event 2983 =====
type=AVC msg=audit(1782840971.324:2983): apparmor="DENIED" operation="capable"
class="cap" profile="unprivileged_usersns" pid=67391 comm="packet_edit_mem"
capability=12 capname="net_admin"
type=SYSCALL msg=audit(1782840971.324:2983): arch=c000003e syscall=44 success=yes
exit=32 a0=4 a1=4bbb40 a2=20 a3=0 items=0 ppid=67390 pid=67391 auid=1000 uid=1000
gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000 tty=pts0
ses=3 comm="packet_edit_mem" exe="/home/user/Downloads/packet_edit_meme-
main/packet_edit_meme" subj=unprivileged_usersns key=(null)
type=PROCTITLE msg=audit(1782840971.324:2983): proctitle="./packet_edit_meme"
===== audit event 2985 =====
type=SYSCALL msg=audit(1782840971.335:2985): arch=c000003e syscall=59 success=yes
exit=0 a0=7ffd6b2601d0 a1=7ffd6b2602f0 a2=7ffd6b260630 a3=48d02b items=2 ppid=67392
pid=67393 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000
sgid=1000 fsgid=1000 tty=pts0 ses=3 comm="aa-exec" exe="/usr/bin/aa-exec"
subj=unconfined key="apparmor_bypass_path"
type=EXECVE msg=audit(1782840971.335:2985): argc=6 a0="aa-exec" a1="-p"
a2="trinity" a3="--" a4="./packet_edit_meme" a5="--in-profile"
type=CWD msg=audit(1782840971.335:2985):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840971.335:2985): item=0 name="/usr/bin/aa-exec"
inode=524801 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1782840971.335:2985): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840971.335:2985):
proctitle=61612D65786563002D70007472696E697479002D2D002E2F7061636B65745F656469745F6
D656D65002D2D696E2D70726F666696C65
===== audit event 2989 =====
type=AVC msg=audit(1782840971.347:2989): apparmor="DENIED" operation="capable"
class="cap" profile="unprivileged_usersns" pid=67394 comm="packet_edit_mem"
capability=21 capname="sys_admin"
type=SYSCALL msg=audit(1782840971.347:2989): arch=c000003e syscall=257 success=no
exit=-13 a0=ffffff9c a1=48d05a a2=1 a3=0 items=1 ppid=67393 pid=67394 auid=1000
uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000
tty=pts0 ses=3 comm="packet_edit_mem" exe="/home/user/Downloads/packet_edit_meme-
main/packet_edit_meme" subj=trinity//&unprivileged_usersns key=(null)
type=CWD msg=audit(1782840971.347:2989):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840971.347:2989): item=0 name="/proc/self/setgroups"
inode=287707 dev=00:17 mode=0100644 ouid=1000 ogid=1000 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840971.347:2989):
proctitle=2E2F7061636B65745F656469745F6D656D65002D2D696E2D70726F666696C65
===== audit event 2992 =====
type=SYSCALL msg=audit(1782840971.347:2992): arch=c000003e syscall=59 success=yes
exit=0 a0=7ffd6b2601d0 a1=7ffd6b2602f0 a2=7ffd6b260630 a3=48d02b items=2 ppid=67392
pid=67395 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000
sgid=1000 fsgid=1000 tty=pts0 ses=3 comm="aa-exec" exe="/usr/bin/aa-exec"
subj=unconfined key="apparmor_bypass_path"
type=EXECVE msg=audit(1782840971.347:2992): argc=6 a0="aa-exec" a1="-p" a2="chrome"
a3="--" a4="./packet_edit_meme" a5="--in-profile"
type=CWD msg=audit(1782840971.347:2992):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840971.347:2992): item=0 name="/usr/bin/aa-exec"

```

```

inode=524801 dev=08:02 mode=0100755 ousid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1782840971.347:2992): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ousid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840971.347:2992):
proctitle=61612D65786563002D70006368726F6D65002D2D002E2F7061636B65745F656469745F6D6
56D65002D2D696E2D70726F666696C65
===== audit event 2996 =====
type=AVC msg=audit(1782840971.356:2996): apparmor="DENIED" operation="capable"
class="cap" profile="unprivileged_usersns" pid=67396 comm="packet_edit_mem"
capability=21 capname="sys_admin"
type=SYSCALL msg=audit(1782840971.356:2996): arch=c000003e syscall=257 success=no
exit=-13 a0=ffffffff a1=48d05a a2=1 a3=0 items=1 ppid=67395 pid=67396 auid=1000
uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000
tty=pts0 ses=3 comm="packet_edit_mem" exe="/home/user/Downloads/packet_edit_meme-
main/packet_edit_mem" subj=chrome//&unprivileged_usersns key=(null)
type=CWD msg=audit(1782840971.356:2996):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840971.356:2996): item=0 name="/proc/self/setgroups"
inode=287732 dev=00:17 mode=0100644 ousid=1000 ogid=1000 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840971.356:2996):
proctitle=2E2F7061636B65745F656469745F6D656D65002D2D696E2D70726F666696C65
===== audit event 2999 =====
type=SYSCALL msg=audit(1782840971.356:2999): arch=c000003e syscall=59 success=yes
exit=0 a0=7ffd6b2601d0 a1=7ffd6b2602f0 a2=7ffd6b260630 a3=48d02b items=2 ppid=67392
pid=67397 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000
sgid=1000 fsgid=1000 tty=pts0 ses=3 comm="aa-exec" exe="/usr/bin/aa-exec"
subj=unconfined key="apparmor_bypass_path"
type=EXECVE msg=audit(1782840971.356:2999): argc=6 a0="aa-exec" a1="-p"
a2="flatpak" a3="--" a4="./packet_edit_meme" a5="--in-profile"
type=CWD msg=audit(1782840971.356:2999):
cwd="/home/user/Downloads/packet_edit_meme-main"
type=PATH msg=audit(1782840971.356:2999): item=0 name="/usr/bin/aa-exec"
inode=524801 dev=08:02 mode=0100755 ousid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1782840971.356:2999): item=1 name="/lib64/ld-linux-x86-64.so.2"
inode=535661 dev=08:02 mode=0100755 ousid=0 ogid=0 rdev=00:00 nametype=NORMAL
cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PROCTITLE msg=audit(1782840971.356:2999):
proctitle=61612D65786563002D7000666C617470616B002D2D002E2F7061636B65745F656469745F6
D656D65002D2D696E2D70726F666696C65

```

Observation	Before mitigation	After mitigation
packet_edit_meme	visible	visible
ns_create	visible	visible
netlink_socket	visible	visible
aa-exec	visible, trinity profile	visible, trinity/chrome/flatpak profiles
tc action pedit offset out of bounds	visible in the kernel journal	no analogous sequence in the post-mitigation window

AppArmor DENIED sys_admin/net_admin	did not effectively block the variant	AppArmor denials visible
/usr/bin/su as euid=0 in the attempt context	yes	not observed
/usr/bin/dash as uid=0/euid=0 in the attempt context	yes	not observed
whoami/id as root in the attempt context	yes	not observed

Conclusion from the post-mitigation phase: kernel.apparmor_restrict_unprivileged_unconfined=1 limited the tested escalation path variant. Attempts remained visible in the logs, but ended with AppArmor denials and did not lead to an analogous root shell in the collected artifacts.

8. Risk Assessment, Recommendations and Detection

Risk should be assessed as highest on systems where users or low-privileged processes can execute code locally. Bastions, shared servers, developer hosts, CI/CD runners, container hosts, machines with SSH access for multiple users, and application servers where a prior vulnerability may provide a webshell or local code execution are particularly important.

Level	Conditions
Critical / high	Local users, CI/CD runner, bastion, container host, vulnerable application allowing code execution
Medium	Host vulnerable, but without typical interactive users; usersns enabled; act_pedit available
Lower	User namespaces disabled or effectively restricted; act_pedit blocked; no local code execution
Still required	Kernel update, because configuration mitigations do not replace the patch

8.1 Mitigation Recommendations

Priority	Recommendation	Objective
1	Update the kernel to a version containing the fix for	Target mitigation of the

	CVE-2026-46331.	vulnerability
2	Maintain kernel.apparmor_restrict_unprivileged_unconfined=1 on Ubuntu systems where it is available.	Limits the aa-exec/unconfined path used in the tested variant
3	Consider kernel.unprivileged_userns_clone=0 on servers that do not require rootless containers, Flatpak, sandboxes and similar mechanisms.	Stronger limitation of the class of exploits requiring user namespaces
4	Block act_pedit if the mechanism is not operationally required.	Reduces exposure of the net/sched/act_pedit component
5	Limit local code execution and SSH access to critical hosts.	Reduces the likelihood of meeting the local-access condition
6	Implement auditd/SIEM rules monitoring the ns_create -> netlink_socket -> aa-exec -> root_exec sequence.	Behavior detection, not only detection of a vulnerable version

8.2 Detection - Correlation Logic

The strongest detection is not based on a single event, but on the correlation of several behaviors within a short time window. In the tested case, the most important detection chain was as follows: an unprivileged user creates a namespace, uses a netlink socket, runs aa-exec to an AppArmor profile, and then processes with euid=0 appear. Kernel entries related to tc action pedit offset out of bounds are an additional indicator.

Logical rule 1. Sequence indicating successful local privilege escalation

```
High-confidence condition:
audit>=1000 AND ns_create
AND netlink_socket
AND apparmor_bypass_path OR aa-exec
AND root_exec with euid=0
AND exe IN (/usr/bin/su, /usr/bin/dash, /usr/bin/whoami, /usr/bin/id, /bin/sh)
AND time_window <= 120s
```

Logical rule 2. Exploitation attempt restricted by AppArmor

```
Post-mitigation attempt condition:
aa-exec with profile IN (trinity, chrome, flatpak, busybox)
AND packet_edit_meme OR unknown local binary
AND AppArmor DENIED capname IN (sys_admin, net_admin)
AND profile CONTAINS unprivileged_userns
```

Logical rule 3. Kernel journal artifact specific to the tested variant

```
Kernel condition:
kernel message CONTAINS "tc action pedit offset"
AND kernel message CONTAINS "out of bounds"
```

8.3 Example auditd Rules

The following rule set corresponds to the configuration used in the test. The rules focus on detecting behavior, not on detecting a specific exploit name.

Extract 16. auditd rules to use as a detection baseline

```
-a always,exit -F arch=b64 -S clone,unshare,setns,clone3 -F auid>=1000 -F auid!=-1
-F key=ns_create
-a always,exit -F arch=b64 -S init_module,delete_module,finit_module -F auid>=1000
-F auid!=-1 -F key=kernel_module
-a always,exit -F arch=b64 -S socket -F a0=0x10 -F auid>=1000 -F auid!=-1 -F
key=netlink_socket
-a always,exit -F arch=b64 -S execve -F euid=0 -F auid>=1000 -F auid!=-1 -F
key=root_exec
-w /usr/bin/aa-exec -p x -k apparmor_bypass_path
-w /usr/bin/unshare -p x -k ns_tools
-w /usr/bin/nsenter -p x -k ns_tools
-w /usr/sbin/tc -p x -k tc_exec
-w /sbin/tc -p x -k tc_exec
-w /usr/sbin/modprobe -p x -k modprobe_exec
-w /sbin/modprobe -p x -k modprobe_exec
-w /usr/bin/kmod -p x -k modprobe_exec
-w /usr/bin/su -p x -k suid_exec
-w /usr/bin/sudo -p x -k suid_exec
-w /bin/bash -p x -k shell_exec
-w /bin/sh -p x -k shell_exec
```

8.4 Example Detection Queries

Example 1. Splunk SPL - auditd correlation

```
index=linux_audit (key="ns_create" OR key="netlink_socket" OR
key="apparmor_bypass_path" OR key="root_exec")
| bin_time span=2m
| stats values(key) as keys values(exe) as exe values(comm) as comm values(auid) as
auid values(euid) as euid by host_time
| search keys="ns_create" keys="netlink_socket" keys="apparmor_bypass_path"
keys="root_exec" euid="0"
```

Example 2. Splunk SPL - kernel journal

```
index=linux_kernel ("tc action pedit offset" AND "out of bounds")
| table _time host message
```

Example 3. XQL - processes relevant to the sequence

```
dataset = xdr_data
| filter event_type = ENUM.PROCESS and action_process_image_path in ("/usr/bin/aa-
exec", "/usr/bin/su", "/usr/bin/dash", "/usr/bin/whoami", "/usr/bin/id")
| fields agent_hostname, action_process_image_path,
action_process_image_command_line, actor_effective_username,
actor_process_image_path, event_timestamp
```

Example 4. XQL - aa-exec launching a suspicious binary

```
dataset = xdr_data
| filter event_type = ENUM.PROCESS and action_process_image_command_line contains
"aa-exec" and action_process_image_command_line contains "packet_edit_meme"
| fields agent_hostname, action_process_image_path,
action_process_image_command_line, actor_effective_username, event_timestamp
```

8.5 Indicators to Monitor

Source	Signal	Meaning
auditd	key=ns_create	Namespace creation/switching by an unprivileged user
auditd	key=netlink_socket	Use of netlink, relevant for tc/net/sched
auditd	key=apparmor_bypass_path	Execution of /usr/bin/aa-exec
auditd	key=root_exec	Process with euid=0 assigned to the auid of a regular user
auditd	exe=/usr/bin/su or exe=/usr/bin/dash	Potential root shell or escalation stage
kernel journal	tc action pedit offset out of bounds	Artifact directly indicating act_pedit
AppArmor	DENIED capname=sys_admin or capname=net_admin	Attempt to obtain capability inside a user namespace
AppArmor	subj=trinity//&unprivileged_usersns or subj=chrome//&unprivileged_usersns	Profile stacking after mitigation

8.6 Limitations and Final Notes

The lack of successful escalation after mitigation means that the tested variant was limited in this environment and with this configuration. It does not formally rule out all possible exploitation variants. AppArmor mitigation is a risk-reduction measure, while the target action remains a kernel update.

Extract 17. Evidence bundle SHA256

```
b044608b1d423905633f49e250f7443f4ce0d4c254096cf5ec55a0208d6fb181 /home/user/pedit-
audit-full-fixed-2026-06-30_174437.tar.gz
```

8.7 Sources

ID	Source	Address / file
[1]	NVD, CVE-2026-46331 Detail	https://nvd.nist.gov/vuln/detail/CVE-2026-46331
[2]	Ubuntu Security, CVE-2026-46331	https://ubuntu.com/security/CVE-2026-46331
[3]	Ubuntu Discourse, Understanding AppArmor User Namespace Restriction	https://discourse.ubuntu.com/t/understanding-apparmor-user-namespace-restriction/58007
[4]	Qualys, Three bypasses of Ubuntu unprivileged user namespace restrictions	https://www.qualys.com/2025/three-bypasses-of-Ubuntu-unprivileged-user-namespace-restrictions.txt
[5]	The Hacker News, New Linux pedit COW Exploit Enables Root Access by Poisoning Cached Binaries	https://thehackernews.com/2026/06/new-linux-pedit-cow-exploit-enables.html
[6]	Red Hat, CVE-2026-46331	https://access.redhat.com/security/cve/cve-2026-46331